



Name: _____
Class: _____
Date: _____

Mr. Rawson • GCSE Physics

P2 Electricity: Test Review Checklist

Test: Monday 5 October, Period 6 — last lesson before half term

Everything on the test is on this page. Tick a box when you can do it without looking at your notes.

1. Circuits, current and resistance

- ☐ Draw and read circuit diagrams using the symbols for a switch, cell, battery, resistor, variable resistor, lamp, fuse, diode, LED, ammeter, voltmeter, thermistor and LDR. Ammeter in **series**, voltmeter in **parallel**.
- ☐ Current is the rate of flow of charge, $Q = It$. It needs a source of potential difference (voltage), and it is the same at every point in a single loop.
- ☐ Use and rearrange $V = IR$: for the same potential difference (voltage), more resistance means less current.

2. Components and I-V graphs

- ☐ Sketch and explain the I-V graphs of a resistor (ohmic: a straight line through the origin), a filament lamp (resistance rises as it heats) and a diode (current one way only). Find resistance at a point: $R = V \div I$.
- ☐ Thermistor: resistance falls as temperature rises (a thermostat). LDR: resistance falls as light increases (lights that switch on in the dark).
- ☐ **RP16**: the circuit that measures the I-V characteristics of a resistor, a filament lamp and a diode, with an ammeter and a voltmeter.

3. Series and parallel

- ☐ Series: the same current everywhere, the supply potential difference (voltage) is shared, and $R_{\text{total}} = R_1 + R_2$. Calculate current, p.d. and resistance.
- ☐ Parallel: the same p.d. across each branch, branch currents add up to the total, and total resistance is less than the smallest resistor. Explain why series raises total resistance and parallel lowers it.
- ☐ **RP15**: how the length of a wire affects its resistance, and resistors combined in series and in parallel.

4. Mains electricity

- ☐ Direct p.d. stays in one direction; alternating p.d. keeps reversing. UK mains is ac, about **230 V** at **50 Hz**.
- ☐ Three-core cable: live (brown, about 230 V), neutral (blue, about 0 V) and earth (green and yellow, 0 V, carries current only in a fault), and the job of each.
- ☐ Explain why the live wire is dangerous even with the switch off, and why any connection between live and earth is dangerous.

5. Power and energy

- ☐ Use $P = VI$ and $P = I^2 R$.
- ☐ Use $E = Pt$ and $E = QV$. The energy an appliance transfers depends on its power and how long it is on.
- ☐ Describe how appliances transfer energy (to a motor's kinetic energy, or to heating), and link an appliance's power rating to the energy it transfers.

6. The National Grid

- ☐ The National Grid is cables and transformers: step-up transformers raise the p.d. for the cables, step-down transformers lower it for homes.
- ☐ Explain why the Grid is efficient: a higher p.d. means a smaller current, so less heating of the cables ($P = I^2 R$).
- ☐ **Higher tier**: use $V_p I_p = V_s I_s$ (given on the equation sheet).

Equations to learn by heart: $Q = It$ $V = IR$ $P = VI$ $P = I^2 R$ $E = Pt$ $E = QV$
Given on the equation sheet (Higher tier): $V_p I_p = V_s I_s$ Practise with the P2 Review Packet.